1.1.1

(b) The soup is not cold

(f) Not a proposition

(g) The sky is not purple

1.1.2

(b)
$t \wedge m$

(d)
$\neg m$

(f)
$\neg t$

1.1.3

(b)
False

(d)
True

(f)

True

1.2.1

(b)
True

(d)
True

1.2.2

(b)
$p \wedge \neg q$

(F)
$r \wedge p \wedge \neg q$

1.2.3

(b)
True

(F)
False

1.2.4

(b) $\neg(p \vee q)$

p	q	$p \vee q$	$\neg(p \vee q)$
T	T	T	F
T	F	T	F
F	T	T	F
F	F	F	T

(d) $(r \vee p) \wedge (\neg r \vee \neg q)$

p	q	r	$(r \vee p) \wedge (\neg r \vee \neg q)$
T	T	T	$(T \vee T) \wedge (F \vee F) = F$
T	T	F	$(F \vee T) \wedge (T \vee F) = T$
T	F	T	$(T \vee T) \wedge (F \vee T) = T$
T	F	F	$(F \vee T) \wedge (T \vee T) = T$
F	T	T	$(T \vee F) \wedge (F \vee F) = F$
F	T	F	$(F \vee F) \wedge (T \vee F) = F$
F	F	T	$(T \vee F) \wedge (F \vee T) = T$
F	F	F	$(F \vee F) \wedge (T \vee T) = F$

1.2 5

(b) $p \wedge q$

I am going to a movie and a party tonight

(d) $\neg p \vee \neg q$

I am neither going to a movie tonight nor going to a party tonight.

1.2.7

(b)

$(B \wedge D) \vee (D \wedge M) \vee (B \wedge M)$

1.3.1

(b)

False

(d)

True

1.3.2

(b) IF he trained for the race, then he finished the race.

Converse: IF he finished the race, then he had trained for the race

Inverse: IF he did not train for the race then he did not finish the race.

Contra positive: IF he did not finish the race then he did not train for the race.

(d) IF it was sunny, then the game was held.

Converse: if the game was held then it was Sunny.
Inverse: if it was not Sunny, Then the game was not held
Contrapositive: If the game was not held, then it was not Sunny.

1.3.4

(b) $(p \to q) \to (q \to p)$

p	q	$(p \to q)$	$(q \to p)$	$(p \to q) \to (q \to p)$
T	T	T	T	T
T	F	F	T	T
F	T	T	F	F
F	T	T	T	T

1.3.5

(b) IF I return to College, Then I won't get a Job
$C \to \neg j$

(f) I will get a Job and return to College
$C \wedge j$

1.3.6

(b) Maintaining a B avarage is necessary for Joe to be eligible for the honors program.
Joe is eligible for the honors program if and

only if he maintains a B average.

1.3.8

(d) $h \rightarrow (a \vee w)$

If traffic was heavy then wither there was accident or roads were wet.

1.4.1

(b) Contradiction

$(p \rightarrow q) \leftrightarrow (p \wedge \neg q)$

p	q	$(p \rightarrow q) \leftrightarrow (p \wedge \neg q)$
F	F	F
F	T	F
T	F	F
T	T	F

(d) tautology

$(p \rightarrow q) \vee p$

p	q	$(p \rightarrow q) \vee p$
F	F	T
F	T	T
T	F	T
T	T	T

1.4.4

(b) Not logically equivalent

$\neg(p \vee \neg q)$ and $\neg p \wedge \neg q$

p	q	$\neg(p \vee \neg q)$
F	F	F
F	T	T
T	F	F
T	T	F

p	q	$\neg p \wedge \neg q$
F	F	T *
F	T	F *
T	F	F
T	T	F

1.5.1

(b) $(p \to q) \wedge (q \wedge)$

$\neg(\neg p \vee q) \vee (p \wedge q)$ = Conditional identity

$(\neg\neg p \wedge \neg q) \vee (p \wedge q)$ = De Morgan's Law

$(p \wedge \neg q) \vee (p \wedge q)$ = Double negation Law

$p \wedge (\neg q \vee q)$ = Distributive law

$p \wedge T$ = Complement law

p = Identity law

1.5.2

(b) $p \wedge (\neg p \to q) = p$

$p \wedge (\neg\neg p \vee q)$ = Conditional identity

$p \wedge (p \vee q)$ = Dougle negation law

p = Absorption Law

1.5.3

(a) $p \to (r \to p)$

$\neg p \vee (r \to p)$ = Conditional identity

$\neg p \vee (\neg r \vee p)$ = Conditional identity

$\neg p \vee (p \vee \neg r)$ = Commutative law

$(\neg p \vee p) \vee \neg r$ = Associative law

$T \vee \neg r$ = Complement LAW

T = Domination LAW

1.6.1

(b) $\neg P(3)$

Proposition. $P(3)$ is TRUE because $P(3)$ is FALSE

(d) $T(s, x)$

NOT a proposition because x is a variable.

(h) $T(s, 16) \rightarrow E(6, 3, 36)$

Proposition. $T(s, 16) \rightarrow E(6, 3, 36)$ is True because the hypothesis of the conditional operation, $T(s, 16)$ is False

1.6.2

(b) $\exists x\,(x + 2 = 1)$

True. Example $x = -1$

(d) $\forall x\,(x^2 - x \neq 0)$

False. Counterexample: $x = 1$ or $x = 0$.

1.6.3

(b) The square of every number is at least 0

$\forall x (x^2 \geq 0)$

(d) Every number is less than or equal to its square
$\forall x (x \leq x^2)$

1.7.1

(b) Everyone got a large bonus
$\forall x B(x)$

(d) Someone who is not on the executive team received a large bonus
$\exists x (\neg T(x) \wedge B(x))$

1.7.2

(b) All students enrolled in the class took the test
$\forall x (E(x) \rightarrow T(x))$

(d) At least one student who is enrolled in the class did not take the test.
$\exists x (E(x) \wedge \neg T(x))$.

1.7.6

(b) $\exists x (E(x) \wedge \neg W(x))$

1.7.7

(b) $\forall x (\neg(x \neq \text{SAM}) \rightarrow W(x))$

(f) $\forall x (\neg M(x) \rightarrow D(x))$

1.8.1

(b) $\neg \exists x (P(x) \vee Q(x))$

$\forall x (\neg P(x) \wedge \neg Q(x))$

(d) $\neg \forall x (P(x) \wedge (Q(x) \vee R(x)))$

$\exists x (\neg P(x) \vee (\neg Q(x) \wedge \neg R(x)))$

1.8.2

(b) Every patient was given the medication or the placebo or both

$\forall x (D(x) \vee P(x))$

Negation: $\neg \forall x (D(x) \vee P(x))$

Applying De Morgan's Law: $\exists x (\neg D(x) \wedge \neg P(x))$

English: There is a patient who was not given the medication and not given the placebo.

(d) Every patient who took the placebo had migraines

$\forall x (P(x) \rightarrow M(x))$

Negation: $\neg \forall x (P(x) \rightarrow M(x))$

Applying De Morgan's law: $\exists x (P(x) \wedge \neg M(x))$
English: Some patient took the placebo and did not have migraines.

1.9.1

(c) $\forall y \, M(x,y)$

NOT a proposition

(d) $\exists x \, \exists y \, M(x,y)$
True

1.9.4

(b) $\forall x \, \exists y \, (P(x,y) \wedge Q(x,y))$
$\exists x \, \forall y \, (\neg P(x,y) \vee \neg Q(x,y))$.